

5. PROJECT DEVELOPMENT PHASE

The Project Development Phase is the implementation stage of FitBuddy. During this phase, the planned architecture and requirements are converted into an operational software application.

5.1 Development Environment

The project is developed using Python 3.11 in Visual Studio Code. A virtual environment is created to manage project-specific dependencies.

The main backend technologies are:

- FastAPI.
- SQLAlchemy.
- SQLite.
- Pydantic.
- Jinja2.
- Uvicorn.

Gemini is integrated for AI-based content generation.

5.2 Backend Implementation

FastAPI is used to create the backend application. It handles requests from the frontend and communicates with the AI and database layers.

The application includes routes for:

/

/generate-workout

/submit-feedback

/view-all-users

/api/health

/api/generate-workout

/api/submit-feedback

The backend validates user information before sending it to the AI service.

5.3 AI Workout Generation

When the user submits their information, the backend constructs an AI prompt containing information such as:

Username

Age

Weight

Fitness Goal

Workout Intensity

The information is passed to Gemini. The model generates a seven-day workout plan.

The generated plan contains:

- Daily focus.
- Warm-up.
- Exercises.
- Sets and repetitions or duration.
- Rest periods.
- Cooldown.
- Safety guidance.

5.4 Nutrition and Recovery Generation

FitBuddy also generates a nutrition or recovery tip according to the user's selected fitness goal.

The system avoids presenting the generated information as medical diagnosis or treatment.

5.5 Feedback-Based Plan Modification

After viewing the generated plan, the user can submit feedback.

For example:

Add more cardio exercises and include another recovery day.

The feedback is processed by the AI service along with the original plan. The system generates an updated seven-day plan.

5.6 Database Integration

SQLAlchemy is used to communicate with SQLite.

The system stores:

- User information.
- Original workout plans.
- Updated workout plans.
- User feedback.
- Nutrition tips.
- Creation and update timestamps.

This allows the application to retain user activity.

5.7 Administrator Module

An administrator can access the dashboard using authentication credentials.

The dashboard provides information about:

- Users.
- Age.
- Weight.
- Goals.
- Workout intensity.
- Original workout plans.
- Updated plans.
- Feedback.

The administrator can also remove stored users.

5.8 Frontend Development

The frontend uses Jinja2 templates.

The main interface provides a form where users enter their fitness information. After processing, the generated plan is displayed in a structured format.

CSS is used to create a modern, responsive interface that works on desktop and mobile screens.

